



Assignments

Recursion – 2

Q1. Print all the elements of an array in reverse order.

Q2. Print index of a given element in an array. If not present, print -1.

Q3. A function countAndSay is defined as:

`countAndSay(1) = "1"`

`countAndSay(n)` is the way you would "say" the digit string from `countAndSay(n-1)`, which is then converted into a different digit string.

So, if sample input is $n = 4$,

`countAndSay(1) = 1`

`countAndSay(2) = "one 1" => 11`

`countAndSay(3) = "two 1" => 21`

`countAndSay(4) = "one 2 one 1" => 1211`

Q4. Given an array of integers, print a sum triangle using recursion from it such that the first level has all array elements. After that, at each level the number of elements is one less than the previous level and elements at the level will be the sum of consecutive two elements in the previous level.

So, if sample input is `[5, 4, 3, 2, 1]`, sample output will be:

`[5, 4, 3, 2, 1]`

`[9, 7, 5, 3]`

`[16, 12, 8]`

`[28, 20]`

`[48]`

Note:- Please try to invest time doing the assignments which are necessary to build a strong foundation. Do not directly Copy Paste using Google or ChatGPT. Please use your brain.